

Workshop 3: solutions

1. (a) The design matrix and parameter vector, ignoring identifiability issues, are

$$\begin{pmatrix} 1 & 1 & 0 & 1 & 0 & x_1 \\ 1 & 1 & 0 & 1 & 0 & x_2 \\ 1 & 1 & 0 & 0 & 1 & x_3 \\ 1 & 1 & 0 & 0 & 1 & x_4 \\ 1 & 0 & 1 & 1 & 0 & x_5 \\ 1 & 0 & 1 & 1 & 0 & x_6 \\ 1 & 0 & 1 & 0 & 1 & x_7 \\ 1 & 0 & 1 & 0 & 1 & x_8 \end{pmatrix} \begin{pmatrix} \alpha \\ \nu_1 \\ \nu_2 \\ \gamma_1 \\ \gamma_2 \\ \beta \end{pmatrix}.$$

This model is not identifiable, since the sum of the second and third columns gives the first column and, similarly, the sum of the fourth and fifth columns gives also the first column. Thus the design matrix is not full column rank (only four columns are linearly independent).

- (b) One way to make the model identifiable is to choose baseline categories, for example $\nu_1 = 0$ and $\gamma_1 = 0$, so that vegetarian and female act as reference groups, in which case,

$$\begin{pmatrix} 1 & 0 & 0 & x_1 \\ 1 & 0 & 0 & x_2 \\ 1 & 0 & 1 & x_3 \\ 1 & 0 & 1 & x_4 \\ 1 & 1 & 0 & x_5 \\ 1 & 1 & 0 & x_6 \\ 1 & 1 & 1 & x_7 \\ 1 & 1 & 1 & x_8 \end{pmatrix} \begin{pmatrix} \alpha \\ \nu_2 \\ \gamma_2 \\ \beta \end{pmatrix}.$$

This parametrisation is now identifiable because the columns of the design matrix are linearly independent.

2. (a) The model in matrix-vector form is given by

$$\begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{14} \\ y_{21} \\ y_{22} \\ y_{23} \\ y_{24} \\ y_{31} \\ y_{32} \\ y_{33} \\ y_{34} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & z_{11} \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & z_{12} \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & z_{13} \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & z_{14} \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & z_{21} \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & z_{22} \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & z_{23} \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & z_{24} \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & z_{31} \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & z_{32} \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & z_{33} \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & z_{34} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \gamma_4 \\ \delta \end{pmatrix} + \begin{pmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{13} \\ \epsilon_{14} \\ \epsilon_{21} \\ \epsilon_{22} \\ \epsilon_{23} \\ \epsilon_{24} \\ \epsilon_{31} \\ \epsilon_{32} \\ \epsilon_{33} \\ \epsilon_{34} \end{pmatrix}$$

The design matrix only has seven linearly independent columns.

- (b) A set of possible identifiability constraints is $\beta_1 = \gamma_1 = 0$, leading to

$$\begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{14} \\ y_{21} \\ y_{22} \\ y_{23} \\ y_{24} \\ y_{31} \\ y_{32} \\ y_{33} \\ y_{34} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & z_{11} \\ 1 & 0 & 0 & 1 & 0 & 0 & z_{12} \\ 1 & 0 & 0 & 0 & 1 & 0 & z_{13} \\ 1 & 0 & 0 & 0 & 0 & 1 & z_{14} \\ 1 & 1 & 0 & 0 & 0 & 0 & z_{21} \\ 1 & 1 & 0 & 1 & 0 & 0 & z_{22} \\ 1 & 1 & 0 & 0 & 1 & 0 & z_{23} \\ 1 & 1 & 0 & 0 & 0 & 1 & z_{24} \\ 1 & 0 & 1 & 0 & 0 & 0 & z_{31} \\ 1 & 0 & 1 & 1 & 0 & 0 & z_{32} \\ 1 & 0 & 1 & 0 & 1 & 0 & z_{33} \\ 1 & 0 & 1 & 0 & 0 & 1 & z_{34} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_2 \\ \beta_3 \\ \gamma_2 \\ \gamma_3 \\ \gamma_4 \\ \delta \end{pmatrix} + \begin{pmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{13} \\ \epsilon_{14} \\ \epsilon_{21} \\ \epsilon_{22} \\ \epsilon_{23} \\ \epsilon_{24} \\ \epsilon_{31} \\ \epsilon_{32} \\ \epsilon_{33} \\ \epsilon_{34} \end{pmatrix}$$

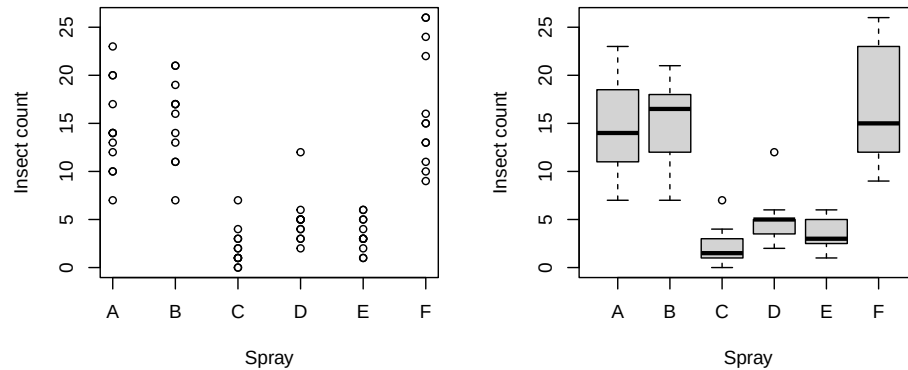
- (c) The model being fitted by the `lm` function involves the main effects of factors `a` and `b` and their interaction, as well as, the effect of the continuous variables `v` and `w` and their product. Its mathematical expression is given by

$$y_i = \mu + \alpha_j + \gamma_k + \delta_{jk} + \beta_1 v_i + \beta_2 w_i + \beta_3 v_i w_i + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2),$$

if observation i is from the j th level of factor `a` and the k th level of factor `b`.

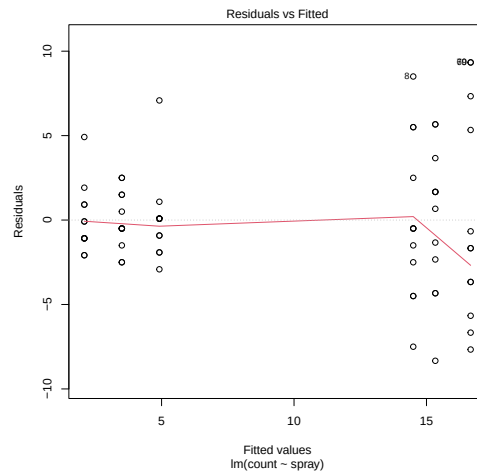
3. (a) The residuals show a clear systematic pattern against x , indicating that x as an effect on y , and this effect is not accounted for in the current model. The model should be extended to include x .
- (b) The residuals versus t show clear serial correlation, violating the independence assumption. This suggests a time-series or correlated-errors model (e.g. including a time trend or using an autoregressive-type error structure) would be more appropriate; such models are beyond the scope of this course, but you should recognise the lack of independence.
4. (a) As we saw in the lecture notes, there are, at least, two possible ways to plot the insect counts against the spray type:

```
par(mfrow = c(2, 2))
stripchart(count ~ spray, data = InsectSprays, vertical = TRUE, pch = 1,
           ylab = "Insect count", xlab = "Spray")
boxplot(count ~ spray, data = InsectSprays,
         ylab = "Insect count", xlab = "Spray")
```

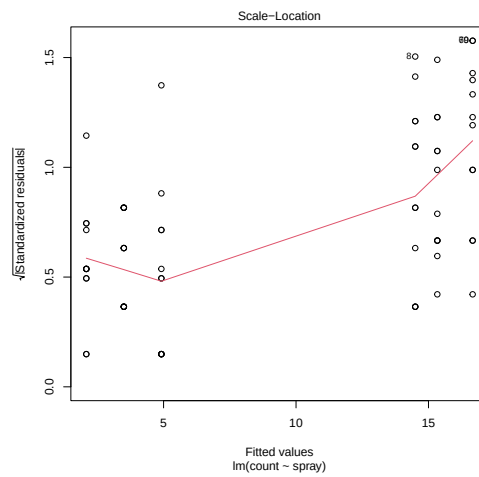


The plot suggests that sprays C, D, and E are the most effective (leading to lower insects counts). Also, note that the variance of the response variable, insect counts, is not constant across the different types of sprays (sprays associated with higher insect counts exhibit greater dispersion in insect counts).

```
(b) im0 <- lm(count ~ spray, data = InsectSprays)
plot(im0, which = 1)
```



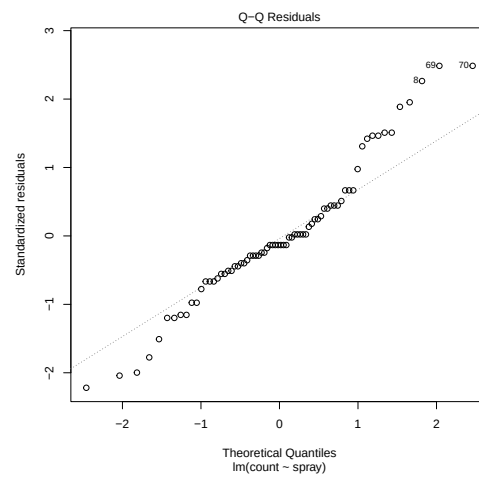
```
plot(im0, which = 3)
```



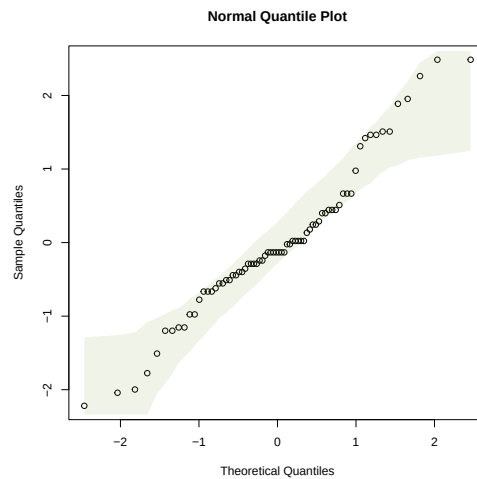
These plots confirm what we observed in (a): the constant variance assumption is not met. In particular, larger fitted values tend to be associated with larger, in absolute value, residuals.

(c) Let us look at the qqplot.

```
plot(im0, which = 2)
library(ecostats)
```



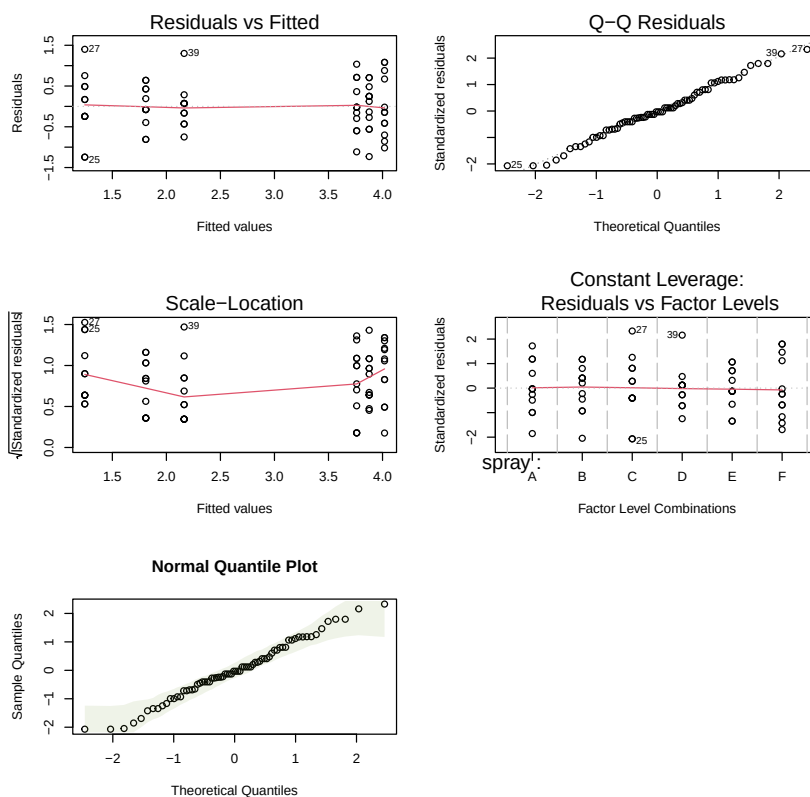
```
qqenvelope(im0)
```



The plot suggests that the normality assumption is not severely violated. There are points departing from the reference line at high and low residuals, suggesting that the residual distribution is ‘heavy tailed’ relative to the normal distribution, but still within the simulation envelope.

(d) Let us fit the model to the square root of the insect counts.

```
im1 <- lm(sqrt(count) ~ spray, data = InsectSprays)
par(mfrow = c(3, 2))
plot(im1)
qqenvelope(im1)
```



The residual plots show substantial improvement, with no remaining evidence against the model assumptions.

(e) Let us look now at the summary of the model in fitted (d).

```
summary(iml)

##
## Call:
## lm(formula = sqrt(count) ~ spray, data = InsectSprays)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.24486 -0.39970 -0.01902  0.42661  1.40089
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.7607     0.1814  20.733 < 2e-16 ***
## sprayB        0.1160     0.2565   0.452   0.653
## sprayC       -2.5158     0.2565  -9.807 1.64e-14 ***
## sprayD       -1.5963     0.2565  -6.223 3.80e-08 ***
## sprayE       -1.9512     0.2565  -7.606 1.34e-10 ***
## sprayF        0.2579     0.2565   1.006   0.318
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6283 on 66 degrees of freedom
## Multiple R-squared:  0.7724, Adjusted R-squared:  0.7552
## F-statistic:  44.8 on 5 and 66 DF,  p-value: < 2.2e-16

head(model.matrix(iml))

##      (Intercept) sprayB sprayC sprayD sprayE sprayF
## 1             1      0      0      0      0      0
## 2             1      0      0      0      0      0
## 3             1      0      0      0      0      0
## 4             1      0      0      0      0      0
## 5             1      0      0      0      0      0
## 6             1      0      0      0      0      0
```

Note that we can write the model as

$$\sqrt{\text{count}_i} = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + \beta_5 x_{5i} + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2),$$

where $x_{1i} = 1$ if spray B was used and $x_{1i} = 0$ otherwise, $x_{2i} = 1$ if spray C was used and $x_{2i} = 0$ otherwise, $x_{3i} = 1$ if spray D was used and $x_{3i} = 0$ otherwise, $x_{4i} = 1$ if spray E was used and $x_{4i} = 0$ otherwise, $x_{5i} = 1$ if spray F was used and $x_{5i} = 0$ otherwise. Spray A is the baseline or reference level, for which $x_{1i} = x_{2i} = x_{3i} = x_{4i} = x_{5i} = 0$. Therefore, for spray A, we have

$$\mathbb{E}(\sqrt{\text{count}_i}) = \beta_0,$$

and so the intercept is the expected square root of the insect count for spray A and this is estimated to be 3.76. When $x_{1i} = 1$ and $x_{2i} = x_{3i} = x_{4i} = x_{5i} = 0$, we have that

$$\mathbb{E}(\sqrt{\text{count}_i}) = \beta_0 + \beta_1,$$

and so β_1 can be interpreted as the expected difference in square root counts between spray B and spray A and this is estimated to be $\hat{\beta}_1 = 0.1160$. The estimated expected square root insect count for spray B is $\hat{\beta}_0 + \hat{\beta}_1 = 3.7607 + 0.1160 = 3.8767$. The remaining parameters have a similar interpretation.

- (f) The previous summary suggested no real differences between sprays A and B and A and F, with C, D, and E all significantly lower than A. Spray C gives the lowest expected square root count, so it might help interpretation to make this the reference level. Let us do that. Note that there is no need to re-check model as it is identical to model `im1`, just with a different parameterisation, so residuals and fitted values are the same as `im1`.

```
InsectSprays$spray <- relevel(InsectSprays$spray, "C")
im2 <- lm(sqrt(count) ~ spray, data = InsectSprays)
summary(im2)
```

```
##
## Call:
## lm(formula = sqrt(count) ~ spray, data = InsectSprays)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-1.24486	-0.39970	-0.01902	0.42661	1.40089

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.2449	0.1814	6.863	2.84e-09 ***
sprayA	2.5158	0.2565	9.807	1.64e-14 ***
sprayB	2.6318	0.2565	10.259	2.67e-15 ***
sprayD	0.9195	0.2565	3.584	0.000641 ***
sprayE	0.5646	0.2565	2.201	0.031238 *
sprayF	2.7738	0.2565	10.813	2.98e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6283 on 66 degrees of freedom
## Multiple R-squared:  0.7724, Adjusted R-squared:  0.7552
## F-statistic: 44.8 on 5 and 66 DF, p-value: < 2.2e-16
```

All other sprays lead to a significant higher number of insect counts than spray C, although D and E are not so very different and at 0.03 the p-value for E is a bit marginal, given that we are conducting six tests at once here (e.g.. if we conduct 20 tests of null hypotheses that are all true we still expect to see one ($= 20 \times 0.05$) p-value below 0.05).

- (g) Let us fit the model, with spray C as the reference level, to the untransformed data.

```

im3 <- lm(count ~ spray, data = InsectSprays)
summary(im3)

##
## Call:
## lm(formula = count ~ spray, data = InsectSprays)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.333 -1.958 -0.500  1.667  9.333
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.083     1.132   1.840  0.0702 .
## sprayA        12.417     1.601   7.755 7.27e-11 ***
## sprayB        13.250     1.601   8.276 8.51e-12 ***
## sprayD         2.833     1.601   1.770  0.0814 .
## sprayE         1.417     1.601   0.885  0.3795
## sprayF        14.583     1.601   9.108 2.79e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.922 on 66 degrees of freedom
## Multiple R-squared:  0.7244, Adjusted R-squared:  0.7036
## F-statistic: 34.7 on 5 and 66 DF,  p-value: < 2.2e-16

```

The differences between sprays D and E and C are now all showing as non-significant. However, given that the constant variance assumption is clearly violated in this case, then the p-values are really not valid. The `im2` results are the more reliable.

5. (a) All predictor variables we will be using are factors. Therefore, let us start by checking that they are correctly coded as such.

```

library(faraway)
is.factor(hsb$gender)

## [1] TRUE

is.factor(hsb$race)

## [1] TRUE

is.factor(hsb$ses)

## [1] TRUE

is.factor(hsb$schtyp)

## [1] TRUE

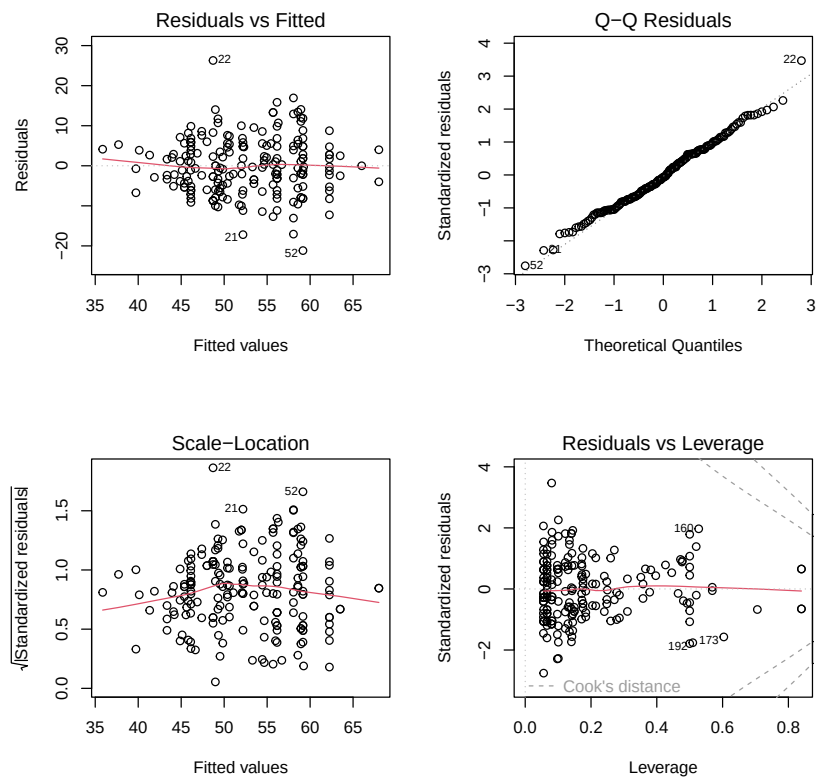
is.factor(hsb$prog)

## [1] TRUE

```


We implement the required model as follows:

```
res_inter <- lm(math ~ (gender + race + ses + schtyp + prog)^2,
                 data = hsb)
par(mfrow = c(2, 2))
plot(res_inter)
```



Everything appears to be fine with the residual plots. The dispersion of residuals is smaller for both low and high fitted values. However, since there are only a few observations in these regions, it is difficult to conclude whether the variance of the error term is constant.

- (b) Let us implement the required backward model selection. To do this automatically, we will use the step function, which performs model selection based on the AIC criterion.

```
step(res_inter, direction = "backward")

## Start:  AIC=862.91
## math ~ (gender + race + ses + schtyp + prog)^2
##
##           Df Sum of Sq   RSS   AIC
## - ses:prog    4   116.58 10042.3 857.25
## - race:schtyp  3    30.99  9956.7 857.54
## - gender:race  3    94.98 10020.7 858.82
## - gender:prog  2     5.65  9931.3 859.03
## - race:ses     6   472.36 10398.0 860.21
## - ses:schtyp   2   101.05 10026.7 860.94
```

```

## - gender:schtyp 1      3.36  9929.1 860.98
## - schtyp:prog 2      134.15 10059.8 861.60
## <none>          9925.7 862.91
## - gender:ses 2      240.56 10166.2 863.70
## - race:prog 6      860.51 10786.2 867.54
##
## Step: AIC=857.25
## math ~ gender + race + ses + schtyp + prog + gender:race + gender:ses +
##       gender:schtyp + gender:prog + race:ses + race:schtyp + race:prog +
##       ses:schtyp + schtyp:prog
##
##           Df Sum of Sq  RSS    AIC
## - race:schtyp 3      33.28 10076 851.91
## - gender:prog 2       2.42 10045 853.30
## - gender:race 3     132.50 10175 853.87
## - ses:schtyp 2      62.44 10105 854.49
## - race:ses 6     480.18 10522 854.59
## - gender:schtyp 1      3.70 10046 855.32
## - schtyp:prog 2     194.69 10237 857.09
## <none>          10042 857.25
## - gender:ses 2     253.63 10296 858.24
## - race:prog 6     779.78 10822 860.20
##
## Step: AIC=851.91
## math ~ gender + race + ses + schtyp + prog + gender:race + gender:ses +
##       gender:schtyp + gender:prog + race:ses + race:prog + ses:schtyp +
##       schtyp:prog
##
##           Df Sum of Sq  RSS    AIC
## - gender:prog 2       2.01 10078 847.95
## - gender:race 3     123.11 10199 848.34
## - ses:schtyp 2      61.62 10137 849.13
## - gender:schtyp 1      1.02 10077 849.93
## - race:ses 6     560.18 10636 850.73
## - schtyp:prog 2     197.49 10273 851.79
## <none>          10076 851.91
## - gender:ses 2     291.52 10367 853.61
## - race:prog 6     821.23 10897 855.58
##
## Step: AIC=847.95
## math ~ gender + race + ses + schtyp + prog + gender:race + gender:ses +
##       gender:schtyp + race:ses + race:prog + ses:schtyp + schtyp:prog
##
##           Df Sum of Sq  RSS    AIC
## - gender:race 3     126.52 10204 844.44
## - ses:schtyp 2      62.18 10140 845.18
## - gender:schtyp 1      1.80 10079 845.99
## - race:ses 6     559.76 10637 846.76

```

```

## - schtyp:prog      2      198.43 10276 847.85
## <none>                                10078 847.95
## - gender:ses       2      305.63 10383 849.93
## - race:prog        6      830.36 10908 851.79
##
## Step:  AIC=844.44
## math ~ gender + race + ses + schtyp + prog + gender:ses + gender:schtyp +
##       race:ses + race:prog + ses:schtyp + schtyp:prog
##
##           Df Sum of Sq  RSS    AIC
## - ses:schtyp      2      85.20 10289 842.11
## - gender:schtyp    1       1.36 10205 842.47
## - race:ses         6     578.45 10782 843.47
## - schtyp:prog      2     192.88 10397 844.19
## <none>                                10204 844.44
## - gender:ses       2     315.99 10520 846.54
## - race:prog        6     783.17 10987 847.23
##
## Step:  AIC=842.11
## math ~ gender + race + ses + schtyp + prog + gender:ses + gender:schtyp +
##       race:ses + race:prog + schtyp:prog
##
##           Df Sum of Sq  RSS    AIC
## - gender:schtyp    1       3.51 10293 840.18
## - race:ses         6     545.29 10835 840.44
## <none>                                10289 842.11
## - gender:ses       2     290.78 10580 843.68
## - schtyp:prog      2     313.59 10603 844.11
## - race:prog        6     813.51 11103 845.33
##
## Step:  AIC=840.18
## math ~ gender + race + ses + schtyp + prog + gender:ses + race:ses +
##       race:prog + schtyp:prog
##
##           Df Sum of Sq  RSS    AIC
## - race:ses         6     542.16 10835 838.44
## <none>                                10293 840.18
## - gender:ses       2     293.72 10586 841.80
## - schtyp:prog      2     310.28 10603 842.12
## - race:prog        6     813.13 11106 843.38
##
## Step:  AIC=838.44
## math ~ gender + race + ses + schtyp + prog + gender:ses + race:prog +
##       schtyp:prog
##
##           Df Sum of Sq  RSS    AIC
## - race:prog        6     492.77 11328 835.34
## <none>                                10835 838.44

```

```
## - gender:ses 2 236.90 11072 838.77
## - schtyp:prog 2 264.78 11100 839.27
##
## Step: AIC=835.34
## math ~ gender + race + ses + schtyp + prog + gender:ses + schtyp:prog
##
##           Df Sum of Sq  RSS    AIC
## - schtyp:prog 2 214.96 11543 835.10
## <none>                11328 835.34
## - gender:ses 2 318.31 11646 836.88
## - race 3 1116.18 12444 848.13
##
## Step: AIC=835.1
## math ~ gender + race + ses + schtyp + prog + gender:ses
##
##           Df Sum of Sq  RSS    AIC
## - schtyp 1 27.18 11570 833.57
## <none>                11543 835.10
## - gender:ses 2 314.23 11857 836.47
## - race 3 1131.77 12674 847.80
## - prog 2 2876.09 14419 875.59
##
## Step: AIC=833.57
## math ~ gender + race + ses + prog + gender:ses
##
##           Df Sum of Sq  RSS    AIC
## <none>                11570 833.57
## - gender:ses 2 301.89 11872 834.72
## - race 3 1125.16 12695 846.13
## - prog 2 2883.26 14453 874.07
##
## Call:
## lm(formula = math ~ gender + race + ses + prog + gender:ses,
##     data = hsb)
##
## Coefficients:
##           (Intercept)                gendermale                raceasian
##           53.7268                -2.0658                8.9369
##           racehispanic                racewhite                seslow
##           0.7998                6.0392                -3.3854
##           sesmiddle                proggeneral                progvocation
##           -3.9952                -6.0894                -9.0475
##           gendermale:seslow  gendermale:sesmiddle
##           0.1996                5.1173
```

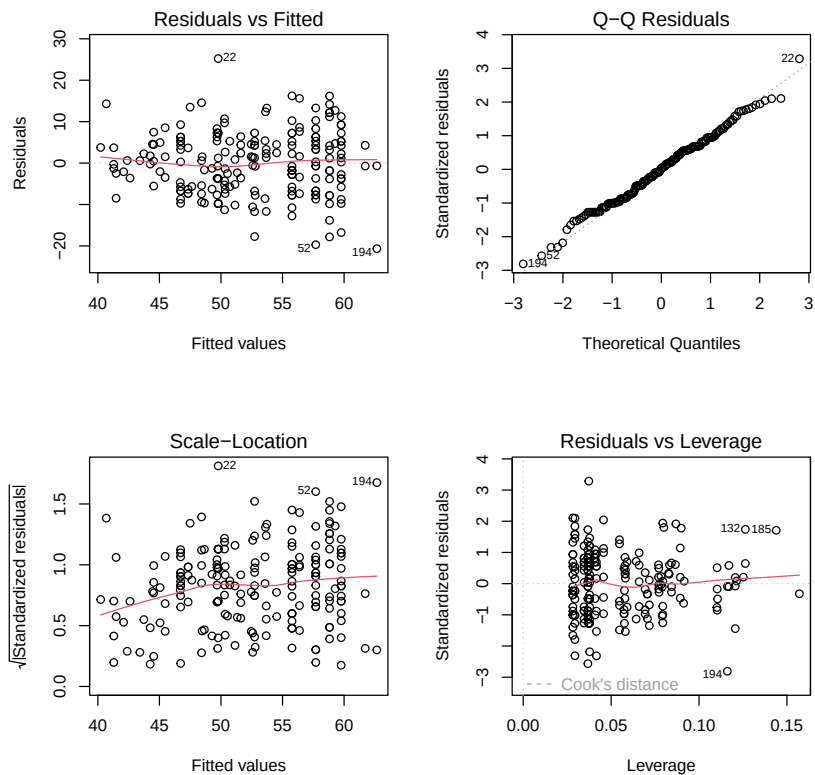
A couple of things to note:

- The model contains ten interaction terms!
- While there are still interactions involving, say, factors u and v , their main effects are never

considered for removal. Just as it has to be.

The final model includes the effects of gender, race, socioeconomic class, high school program, and the interaction between gender and socioeconomic class. Let us fit this model and check the residual plots.

```
res_final <- lm(math ~ gender + race + ses + prog + gender:ses,
                 data = hsb)
par(mfrow = c(2, 2))
plot(res_final)
```



```
summary(res_final)

##
## Call:
## lm(formula = math ~ gender + race + ses + prog + gender:ses,
##     data = hsb)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.6637  -5.7672   0.2553   5.2340  25.2253
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      53.7268     2.3461  22.901  < 2e-16 ***
## gendermale      -2.0658     2.0740  -0.996  0.32050
```

```
## raceasian      8.9369      2.9972      2.982      0.00324 **
## racehispanic   0.7998      2.3973      0.334      0.73904
## racewhite      6.0392      1.9597      3.082      0.00237 **
## seslow        -3.3854      2.0929     -1.618      0.10741
## sesmiddle     -3.9952      1.8749     -2.131      0.03439 *
## proggeneral   -6.0894      1.4372     -4.237      3.54e-05 ***
## progvocation  -9.0475      1.3940     -6.490      7.34e-10 ***
## gendermale:seslow  0.1996      3.2281      0.062      0.95077
## gendermale:sesmiddle 5.1173      2.6221      1.952      0.05246 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.824 on 189 degrees of freedom
## Multiple R-squared:  0.3376, Adjusted R-squared:  0.3025
## F-statistic: 9.631 on 10 and 189 DF,  p-value: 6.52e-13
```

The residual plots do not show any severe violations of the model assumptions. Let us mathematically express this final model. To determine which levels of the factor variables were set as the baseline, one possibility is to examine the model matrix.

```
head(model.matrix(res_final))

##      (Intercept) gendermale raceasian racehispanic racewhite seslow sesmiddle
## 1             1           1           0           0           1           1           0
## 2             1           0           0           0           1           0           1
## 3             1           1           0           0           1           0           0
## 4             1           1           0           0           1           0           0
## 5             1           1           0           0           1           0           1
## 6             1           1           0           0           1           0           1
##  proggeneral progvocation gendermale:seslow gendermale:sesmiddle
## 1             1           0           1           0
## 2             0           1           0           0
## 3             1           0           0           0
## 4             0           1           0           0
## 5             0           0           0           1
## 6             0           0           0           1
```

Thus, the baseline for gender is female, for race is african-american, for socioeconomic class is high, and for high school program is academic. We now define the following dummy or indicator

variables:

$$\begin{aligned}
\text{gender}_{\text{male},i} &= \begin{cases} 1, & \text{if student } i \text{ is male,} \\ 0, & \text{otherwise.} \end{cases} \\
\text{race}_{\text{asian},i} &= \begin{cases} 1, & \text{if student } i \text{ is asian,} \\ 0, & \text{otherwise.} \end{cases} \\
\text{race}_{\text{hispanic},i} &= \begin{cases} 1, & \text{if student } i \text{ is hispanic,} \\ 0, & \text{otherwise.} \end{cases} \\
\text{race}_{\text{white},i} &= \begin{cases} 1, & \text{if student } i \text{ is white,} \\ 0, & \text{otherwise.} \end{cases} \\
\text{ses}_{\text{low},i} &= \begin{cases} 1, & \text{if student } i \text{ is from a low socioeconomic class ,} \\ 0, & \text{otherwise.} \end{cases} \\
\text{ses}_{\text{middle},i} &= \begin{cases} 1, & \text{if student } i \text{ is from a middle socioeconomic class ,} \\ 0, & \text{otherwise.} \end{cases} \\
\text{prog}_{\text{general},i} &= \begin{cases} 1, & \text{if student } i \text{ is in a general high school program ,} \\ 0, & \text{otherwise.} \end{cases} \\
\text{prog}_{\text{vocation},i} &= \begin{cases} 1, & \text{if student } i \text{ is in a vocational high school program ,} \\ 0, & \text{otherwise.} \end{cases}
\end{aligned}$$

We can now write the model as

$$\begin{aligned}
y_i &= \beta_0 + \beta_{\text{male}} \text{gender}_{\text{male},i} + \beta_{\text{asian}} \text{race}_{\text{asian},i} + \beta_{\text{hispanic}} \text{race}_{\text{hispanic},i} + \beta_{\text{white}} \text{race}_{\text{white},i} \\
&+ \beta_{\text{low}} \text{ses}_{\text{low},i} + \beta_{\text{middle}} \text{ses}_{\text{middle},i} + \beta_{\text{general}} \text{prog}_{\text{general},i} + \beta_{\text{vocation}} \text{prog}_{\text{vocation},i} \\
&+ \beta_{\text{male,low}} \text{gender}_{\text{male},i} \text{ses}_{\text{low},i} + \beta_{\text{male,middle}} \text{gender}_{\text{male},i} \text{ses}_{\text{middle},i} + \epsilon_i,
\end{aligned}$$

with $\epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, for $i = 1, \dots, n = 200$. In this model β_0 represents the expected math score for an African-American female, from a high socioeconomic status, who is enrolled in an academic high school program.

- (c) We observe that, among students of the same gender, belonging to the same socioeconomic class, and enrolled in the same high school program, the estimated expected difference in math scores is 8.94 points between Asian and African American students, 0.80 points between Hispanic and African American students, and 6.04 points between white and African American students.

In a similar fashion, among students of the same gender and race and belonging to the same socioeconomic class, the estimated expected difference in math scores is -6.1 points between students enrolled in a general and in an academic high school program and of -9.05 points between students enrolled in a vocational and in an academic high school program.

In turn, any comments regarding the effect of gender on math scores are dependent on the socioeconomic class, and vice versa. For instance, among students of the same race and enrolled in the

same high school program, the estimated expected difference in math scores between a male and a female student, both belonging to a high socioeconomic class (baseline) is -2.07 points. Further, among students of the same race and enrolled in the same high school program, the estimated expected difference in math scores between a male and a female student, both belonging to a low socioeconomic class is $-2.0658 + 0.1996 = -1.87$ points.

6. (a) We begin by examining the variables.

```
str(teengamb)

## 'data.frame': 47 obs. of 5 variables:
## $ sex : int 1 1 1 1 1 1 1 1 1 1 ...
## $ status: int 51 28 37 28 65 61 28 27 43 18 ...
## $ income: num 2 2.5 2 7 2 3.47 5.5 6.42 2 6 ...
## $ verbal: int 8 8 6 4 8 6 7 5 6 7 ...
## $ gamble: num 0 0 0 7.3 19.6 0.1 1.45 6.6 1.7 0.1 ...
```

The variable `sex` is not coded as a factor. Although this does not affect the results here (since it has only two levels), it is good practice to explicitly code factor variables.

```
teengamb$sex <- as.factor(teengamb$sex)
```

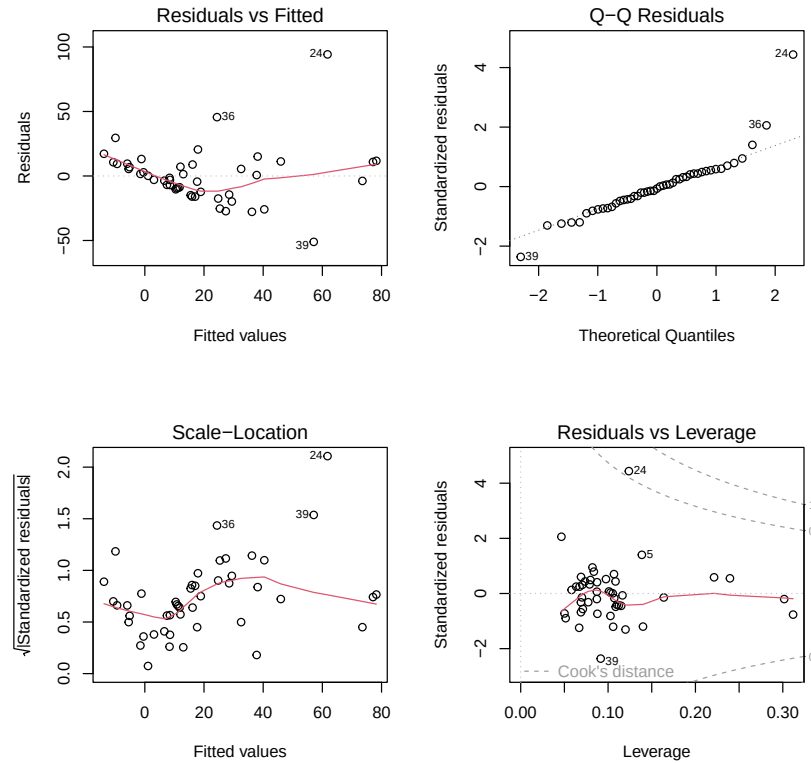
Let us now fit the model with gambling expenditure as the response and `sex`, socioeconomic status, income, and verbal score as predictors. We will then examine the residual plots.

```
fit_gamble <- lm(gamble ~ sex + status + income + verbal,
                 data = teengamb)
summary(fit_gamble)

##
## Call:
## lm(formula = gamble ~ sex + status + income + verbal, data = teengamb)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -51.082 -11.320  -1.451   9.452  94.252
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  22.55565   17.19680   1.312   0.1968
## sex1        -22.11833    8.21111  -2.694   0.0101 *
## status         0.05223    0.28111   0.186   0.8535
## income         4.96198    1.02539   4.839 1.79e-05 ***
## verbal        -2.95949    2.17215  -1.362   0.1803
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22.69 on 42 degrees of freedom
## Multiple R-squared:  0.5267, Adjusted R-squared:  0.4816
## F-statistic: 11.69 on 4 and 42 DF, p-value: 1.815e-06
```



```
par(mfrow = c(2, 2))
plot(fit_gamble)
```



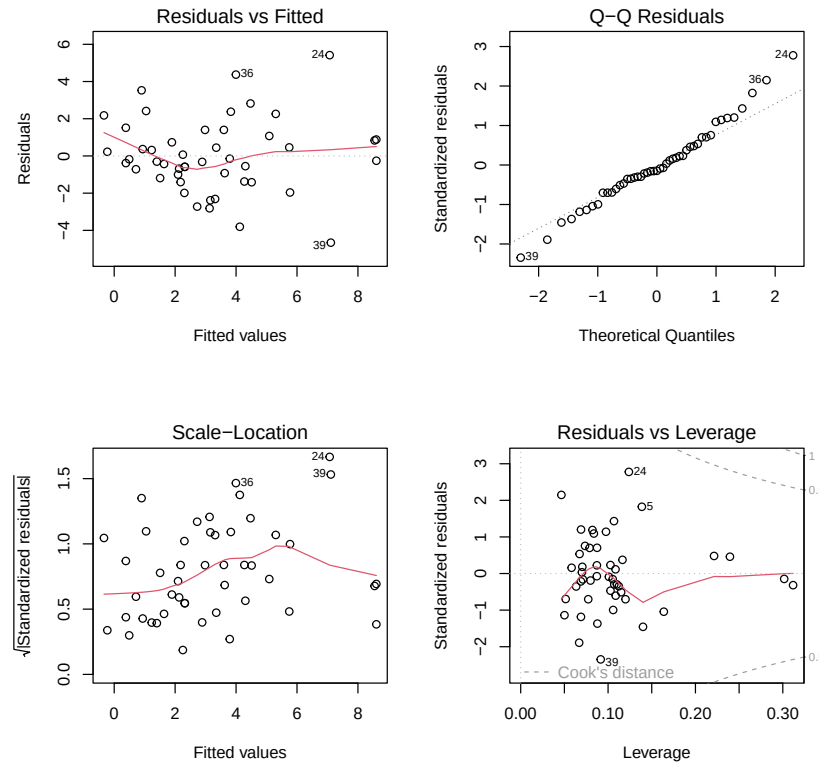
The variance of the residuals seems to increase with the fitted values, indicating non constant variance. Observation 24 also has a large standardized residual and a borderline Cook's distance value. Let us now fit the model with a transformed response variable, i.e., with the square root of gambling expenditure.

```
fit_gamble_sqrt <- lm(sqrt(gamble) ~ sex + status + income + verbal,
                       data = teengamb)
summary(fit_gamble_sqrt)
```

```
##
## Call:
## lm(formula = sqrt(gamble) ~ sex + status + income + verbal, data = teengamb)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.6606 -1.0961 -0.2564  0.9786  5.4178
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.97707    1.57947   1.885  0.06638 .
## sex1         -2.04450    0.75416  -2.711  0.00968 **
## status         0.03688    0.02582   1.428  0.16057
## income         0.47938    0.09418   5.090 7.94e-06 ***
##
```

```
## verbal      -0.42360      0.19950     -2.123     0.03967 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.084 on 42 degrees of freedom
## Multiple R-squared:  0.5646, Adjusted R-squared:  0.5231
## F-statistic: 13.61 on 4 and 42 DF,  p-value: 3.362e-07

par(mfrow = c(2, 2))
plot(fit_gamble_sqrt)
```



Although some heteroscedasticity remains, the square-root transformation of gambling expenditure seems to improve this issue. QQ plot also looks better for the model with transformed response. Observation 24 is no longer ‘problematic’.

(b) Let us first write the model

$$\sqrt{\text{gamble}_i} = \beta_1 + \beta_2 \text{sex}_{F,i} + \beta_3 \text{status}_i + \beta_4 \text{income}_i + \beta_5 \text{verbal}_i + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2),$$

where $\text{sex}_{F,i}$ is a dummy variable taking the value 1 if teenager i is a female and 0 if a male. With all other predictor variables held constant, say at values status^* , income^* , and verbal^* , the estimated difference in the mean (square root) expenditure on gambling for a male compared to

a female is

$$\begin{aligned}
 & (\hat{\beta}_1 + \hat{\beta}_2 \times 0 + \hat{\beta}_3 \times \text{status}^* + \hat{\beta}_4 \times \text{income}^* + \hat{\beta}_5 \times \text{verbal}^*) - \\
 & (\hat{\beta}_1 + \hat{\beta}_2 \times 1 + \hat{\beta}_3 \times \text{status}^* + \hat{\beta}_4 \times \text{income}^* + \hat{\beta}_5 \times \text{verbal}^*) \\
 & = -\hat{\beta}_2 \\
 & = 2.04450
 \end{aligned}$$

Thus, holding status, income, and verbal score fixed, males are estimated to spend on average 2.04 units more than females on the square-root scale. A 95% confidence interval for this difference is given by

$$(-\hat{\beta}_2 \pm t_{n-p, 0.975} \hat{\sigma}_{\hat{\beta}_2}), \quad n - p = 47 - 5 = 42, \quad \hat{\sigma}_{\hat{\beta}_2} = 0.75416.$$

```

lb <- -coef(fit_gamble_sqrt)[2] - qt(0.975, df = 42)*0.75416
ub <- -coef(fit_gamble_sqrt)[2] + qt(0.975, df = 42)*0.75416
lb; ub

##      sex1
## 0.5225457
##      sex1
## 3.566459

```

The required 95% CI is (0.523, 3.566).

Note: If you have obtained the symmetric estimate and interval, that is fine as well.

- (c) We simply need to find the maximum values for a male, which in this case coincide with the maximum values for the dataset.

```

ind_male <- which(teengamb$sex == "0")
df_male <- data.frame(sex = "0",
                      status = max(teengamb$status[ind_male]),
                      income = max(teengamb$income[ind_male]),
                      verbal = max(teengamb$verbal[ind_male])
                      )
predict(fit_gamble_sqrt, newdata = df_male,
        interval = "prediction", level = 0.95)^2

##      fit      lwr      upr
## 1 75.65038 13.80692 187.1336

```

For such a hypothetical observation, the predicted gambling expenditure is £75.65, with a 95% prediction interval for a future expenditure of (£13.81, £187.13).

- (d) Let us start by fitting the model where income is the only predictor, i.e., let us start by fitting the following model:

$$\sqrt{\text{gamble}_i} = \beta_1 + \beta_4 \text{income}_i + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2),$$

where we note that although we are using the same notation, the coefficients estimates of β_1 and β_4 in this model will be different from those in the full model (and their interpretation is also, of course, different!).

```
fit_gamble_sqrt_income <- lm(sqrt(gamble) ~ income, data = teengamb)
```

We are then interested in testing the following hypothesis:

$$H_0 = \begin{pmatrix} \beta_2 \\ \beta_3 \\ \beta_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad H_1 = \begin{pmatrix} \beta_2 \\ \beta_3 \\ \beta_5 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

This hypothesis can be tested through an F-ratio test:

$$F = \frac{(RSS_0 - RSS_1)/(p_1 - p_0)}{RSS_1/(n - p_1)} \sim F_{p_1 - p_0, n - p_1},$$

where RSS_0 is the residual sum of squares of the model including income as the only predictor and RSS_1 is the residual sum of squares of the full model. Here $p_1 - p_0$ is equal to three. We know that in practice, there is no need to obtain all these quantities ‘manually’, and that the `anova` command can be used for this purpose. This is what we will do next.

```
anova(fit_gamble_sqrt_income, fit_gamble_sqrt)

## Analysis of Variance Table
##
## Model 1: sqrt(gamble) ~ income
## Model 2: sqrt(gamble) ~ sex + status + income + verbal
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      45 272.76
## 2      42 182.41   3    90.346 6.9339 0.0006774 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p-value is 0.0006774 and therefore we reject the null hypothesis: that is, at least, one of the coefficients β_2 , β_3 , and β_5 is different than zero (does not necessarily mean that the three coefficients are all different than zero).

- (e) Excluding interactions between variables, there are eight possible models. The estimated effect, 95% CI, and p-value of ‘sex’ across these eight models are as follows.

```
fit_gamble_sqrt_1 <- lm(sqrt(gamble) ~ sex, data = teengamb)
fit_gamble_sqrt_2 <- lm(sqrt(gamble) ~ sex + income, data = teengamb)
fit_gamble_sqrt_3 <- lm(sqrt(gamble) ~ sex + status, data = teengamb)
fit_gamble_sqrt_4 <- lm(sqrt(gamble) ~ sex + verbal, data = teengamb)
fit_gamble_sqrt_5 <- lm(sqrt(gamble) ~ sex + status + income,
  data = teengamb)
fit_gamble_sqrt_6 <- lm(sqrt(gamble) ~ sex + income + verbal,
  data = teengamb)
fit_gamble_sqrt_7 <- lm(sqrt(gamble) ~ sex + status + verbal,
  data = teengamb)
fit_gamble_sqrt_8 <- lm(sqrt(gamble) ~ sex + income + verbal + status,
  data = teengamb)

summary_1 <- summary(fit_gamble_sqrt_1)
```

```

summary_2 <- summary(fit_gamble_sqrt_2)
summary_3 <- summary(fit_gamble_sqrt_3)
summary_4 <- summary(fit_gamble_sqrt_4)
summary_5 <- summary(fit_gamble_sqrt_5)
summary_6 <- summary(fit_gamble_sqrt_6)
summary_7 <- summary(fit_gamble_sqrt_7)
summary_8 <- summary(fit_gamble_sqrt_8)

ci_1 <- confint(fit_gamble_sqrt_1)
ci_2 <- confint(fit_gamble_sqrt_2)
ci_3 <- confint(fit_gamble_sqrt_3)
ci_4 <- confint(fit_gamble_sqrt_4)
ci_5 <- confint(fit_gamble_sqrt_5)
ci_6 <- confint(fit_gamble_sqrt_6)
ci_7 <- confint(fit_gamble_sqrt_7)
ci_8 <- confint(fit_gamble_sqrt_8)

df <- data.frame("Estimate" = c(summary_1$coefficients[2, 1],
                                summary_2$coefficients[2, 1],
                                summary_3$coefficients[2, 1],
                                summary_4$coefficients[2, 1],
                                summary_5$coefficients[2, 1],
                                summary_6$coefficients[2, 1],
                                summary_7$coefficients[2, 1],
                                summary_8$coefficients[2, 1]),
                 "LB" = c(ci_1[2, 1],
                           ci_2[2, 1],
                           ci_3[2, 1],
                           ci_4[2, 1],
                           ci_5[2, 1],
                           ci_6[2, 1],
                           ci_7[2, 1],
                           ci_8[2, 1]),
                 "UB" = c(ci_1[2, 2],
                           ci_2[2, 2],
                           ci_3[2, 2],
                           ci_4[2, 2],
                           ci_5[2, 2],
                           ci_6[2, 2],
                           ci_7[2, 2],
                           ci_8[2, 2]),
                 "Pval" = c(summary_1$coefficients[2, 4],
                             summary_2$coefficients[2, 4],
                             summary_3$coefficients[2, 4],
                             summary_4$coefficients[2, 4],
                             summary_5$coefficients[2, 4],
                             summary_6$coefficients[2, 4],

```

Table 1: Question 2(e): estimate of sex coefficient, 95 CI, and p-value.

	Estimate	Lower Bound 95 CI	Upper Bound 95 CI	P-value
Sex	-2.8884	-4.4960	-1.2809	0.0007
Sex + Income	-2.5069	-3.8009	-1.2130	0.0003
Sex + Status	-3.4572	-5.2775	-1.6369	0.0004
Sex + Verbal	-3.0611	-4.6220	-1.5002	0.0003
Sex + Status + Income	-2.3624	-3.9127	-0.8121	0.0037
Sex + Income + Verbal	-2.6389	-3.9226	-1.3553	0.0002
Sex + Status + Verbal	-3.1684	-4.9960	-1.3409	0.0011
Sex + Income + Verbal + Status	-2.0445	-3.5665	-0.5225	0.0097

```
summary_7$coefficients[2, 4],
summary_8$coefficients[2, 4]))

colnames(df) <- c("Estimate", "Lower Bound 95 CI",
  "Upper Bound 95 CI", "P-value")
rownames(df) <- c("Sex", "Sex + Income", "Sex + Status",
  "Sex + Verbal",
  "Sex + Status + Income", "Sex + Income + Verbal",
  "Sex + Status + Verbal",
  "Sex + Income + Verbal + Status")
library(kableExtra)
knitr::kable(df, escape = FALSE, digits = 4,
  caption = "Question 2(e): estimate of sex coefficient,
  95 CI, and p-value.") %>%
  kable_styling(position = "center")
```

As shown in the table below, although the estimated effect ranges from -2.0445 to -3.4572 , the corresponding 95% CIs exhibit good overlap, and the p-value remains below 0.05 for all models. Therefore, the estimated effect of sex appears to be quite stable across different models.

- (f) There is no single right answer here. Not all eight models are nested, so we use AIC for model selection, although all models are nested within the full model that includes all four predictors.

```
AIC(fit_gamble_sqrt_1, fit_gamble_sqrt_2, fit_gamble_sqrt_3,
  fit_gamble_sqrt_4, fit_gamble_sqrt_5, fit_gamble_sqrt_6,
  fit_gamble_sqrt_7, fit_gamble_sqrt_8)

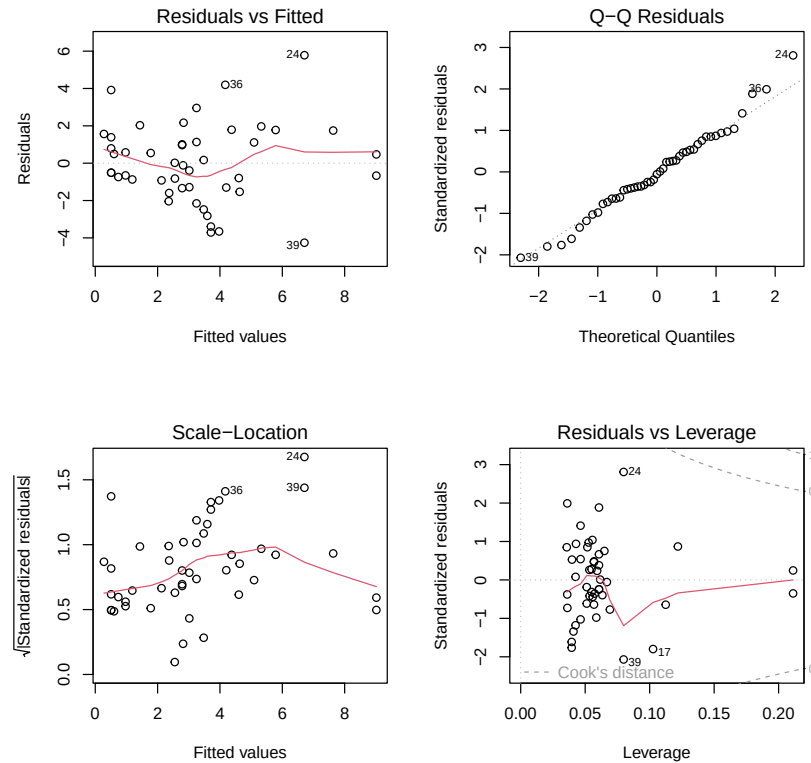
##           df      AIC
## fit_gamble_sqrt_1  3 230.1886
## fit_gamble_sqrt_2  4 210.0445
## fit_gamble_sqrt_3  4 230.3917
## fit_gamble_sqrt_4  4 227.7623
## fit_gamble_sqrt_5  5 211.9106
## fit_gamble_sqrt_6  5 209.3479
## fit_gamble_sqrt_7  5 229.7019
## fit_gamble_sqrt_8  6 209.1185
```

The AIC is lowest for the full model containing all four predictors. However, model 6 (which includes sex, income, and verbal score), model 5 (which includes sex, socioeconomic status, and income), and model 2 (which includes only sex and income) have almost identical AIC values. Given the small sample size ($n = 47$), the more parsimonious model 2 is appealing (which has an adjusted r^2 of 0.49 compared to 0.52 for the full model). Let us analyse the summary of this model and inspect the residual plots.

```
summary(fit_gamble_sqrt_2)

##
## Call:
## lm(formula = sqrt(gamble) ~ sex + income, data = teengamb)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.2597 -1.2946 -0.1159  1.2604  5.7808
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.09425     0.60297   3.473 0.001166 **
## sex1          -2.50694     0.64205  -3.905 0.000321 ***
## income         0.46149     0.08968   5.146 5.95e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.146 on 44 degrees of freedom
## Multiple R-squared:  0.5164, Adjusted R-squared:  0.4945
## F-statistic: 23.5 on 2 and 44 DF, p-value: 1.142e-07

par(mfrow = c(2, 2))
plot(fit_gamble_sqrt_2)
```



The residual diagnostics are not noticeably worse than those for the full model. The estimated effect of sex is -2.51 , meaning that, at the same income level, females spend less on gambling than males (on the square root scale). The estimated income effect is 0.46 , meaning that, keeping gender fixed, each pound increase (per week) leads to an additional expenditure on gambling of 0.46 (in square root scale). In a more meaningful way, a $\pounds 10$ increase in income leads to a $\pounds 4.6$ increase of (square root) expenditure on gambling.